Unit 4 - Important Topics

Feed systems in liquid propellant rockets aur problems

21ASC305T - Rocket Propulsion
Slides ke basis par simple Hinglish me samjhaya hua notes

Unit 4 - Important Topics (Liquid Rocket)

Ye PDF sirf important topics ka hai: liquid propellant rockets ke feed systems aur liquid wale problems.

Is PDF me kya kya hai

- Feed systems in liquid propellant rockets: gas pressure feed (regulated aur blow-down), turbopump se comparison, pressurizing gas sources
- Problems: propellant mass, exhaust velocity, specific impulse, chamber pressure, thrust (liquid engine ke liye)

Source ka matlab: hare box me PDF ka naam aur page number (slide number) diya hai. Feed systems wala poora part slides se hai (UNIT 4 RP - L2.pdf, Page 4 se 20). Problems ke formulas Unit 3 ke slides (thrust, C*, I_{sp}, mass flow) se hain; liquid wale numericals repo ke slides me nahi the, isliye maine practice ke liye banaye hain.

1. Propellant feed systems

Source: UNIT 4 RP - L2.pdf | Page 4 se 8 (functions p4, selection p5, gas vs turbopump p6, diagram p7, turbopump tank pressure p8)

Feed system ke 2 principal functions hain:

- Propellants ka pressure badhana.
- Unhe design mass flow rate par ek ya zyada thrust chambers tak supply karna.

In kaamo ki energy ya to **high-pressure gas** se aati hai, ya **centrifugal pumps** se, ya dono ke combination se.

Feed system ka selection kis par depend karta hai

- Rocket application aur mission
- Duration
- Thrust chambers ki number ya type
- Past experience
- Design ki simplicity, manufacturing ki aasani, low cost aur minimum inert mass

Gas pressure feed vs turbopump feed

Gas pressure feed systems turbopump systems se **behtar** performance dete hain jab:

- Vehicle ka total impulse ya propellant mass relatively low ho.
- Chamber pressure relatively low ho.
- Engine thrust-to-weight ratio low ho (usually 0.6 se kam).

Gas pressure system me heavy-walled propellant tanks aur pressurizing gas hi engine system ka major inert mass hote hain.

Turbopump feed system me propellant tank pressure **10 se 40 guna kam** hota hai, aur isliye vehicle ke tank masses bhi usi factor se kam ho jaate hain.

Diagram (Page 7): regulated gas pressure feed system ki schematic. Labels: gas tank, gas start valve, pressure regulator, fuel tank aur oxidizer tank, fuel main valve, oxidizer main valve, injector, combustion chamber, exhaust nozzle, check valves, vent and relief valves, isolation valves, fill and drain valves.

	Gas pressure feed	Turbopump feed
Kab behtar	Low total impulse, low chamber pressure, low thrust-to-weight (< 0.6)	Bade impulse aur high chamber pressure ke liye
Tank pressure	High (tank ko heavy-walled banana padta hai)	10 se 40 guna kam

	Gas pressure feed	Turbopump feed
Tank mass	Zyada	Kaafi kam (usi factor se)
Energy source	High-pressure gas	Centrifugal pumps (turbine driven)

2. Gas pressure feed systems

Source: UNIT 4 RP - L2.pdf | Page 9 se 15 (types p9, regulated components p10, diagram p11, scavenging p12, blow-down p13-14, comparison table p15)

Liquid propellants ko tanks se bahar dhakelne ka sabse simple aur common tareeka high-pressure gas hai. Aaj bhi do common types use hote hain:

Type 1: Regulated gas pressure feed system

- Gas feed line me ek **gas pressure regulator** hota hai. Isse engine lagbhag constant tank pressure par aur lagbhag constant thrust par chalta hai.
- Main components: high-pressure gas tank, gas starting valve, gas pressure regulator, propellant tanks, propellant valves aur feed lines.

Extra components bhi ho sakte hain: filling aur draining provisions, check valves, filters, flexible elastic bladders (liquid ko pressurizing gas se alag rakhne ke liye), pressure sensors ya gauges.

- Jab propellants poore consume ho jaate hain to pressurizing gas ka use lines aur valves se bacha hua liquid propellant residue saaf karne (scavenge) ke liye bhi kar sakte hain.
- Agar unit ko baar baar use ya fly karna ho (jaise space-maneuver rocket) to usme kuch extra features ho sakte hain, jaise thrust-regulating device aur tank level gauge.

Type 2: Blow-down feed system

- Propellant tanks **bade** hote hain kyunki unme propellants ke saath pressurizing gas bhi initial maximum tank pressure par store hoti hai.
- Alag high-pressure gas tank aur pressure regulator **nahi** hote.
- Tank ke andar pehle se bhari gas ka expansion hi propellants ko bahar dhakelta hai. (Isliye tank pressure aur thrust dheere dheere girte jaate hain.)

Diagram (Page 11): regulated system ka detailed layout: high-pressure gas supply tank, gas bleed valve, pressure regulator, tank vent valves, filters, check valves, oxidizer tank aur fuel tank, propellant valves (remote control), restricting orifice, rocket thrust chamber aur optional additional thrust chamber(s).

	Regulated system	Blow-down system
Gas storage	Alag high-pressure gas tank	Propellant tank ke andar hi gas
Regulator	Hota hai	Nahi hota
Tank size	Propellant tanks chhote	Propellant tanks bade
Tank pressure / thrust	Lagbhag constant	Burn ke saath girta hai
Complexity	Zyada parts	Simple

Slide ki comparison table (Page 15)

	Regulated pressure	Blowdown
Pressure / thrust	Essentially constant	Propellant consume hone ke saath decrease hota hai
Gas storage	Alag high-pressure tanks me	Gas propellant tank ke andar hi stored (tank pressure par, large ullage volume 30-60%)
Required components	Regulator, filter, gas valves aur gas tank chahiye	Bade, bhaari propellant tanks
Advantages	Nearly constant-pressure feed se essentially constant propellant flow aur approximately constant thrust, Isp aur mixture ratio. Mixture ratio ka behtar control.	Simpler system. Kam gas chahiye. Inert mass kam ho sakta hai. Koi high-pressure gas tank nahi.
Disadvantages	Thoda zyada complex. Regulator se chhota pressure drop. Gas high pressure par lambe time tak stored rehti hai. Zyada pressurizing gas chahiye.	Thrust burning duration ke saath girta hai. Mixture ratio ka control kam accurate hone se thoda zyada residual propellant. Thruster ko thrust values ki wide range aur mixture ratio ki modest range me stable chalna padta hai. Propellants pressure me stored; burning time ke end par Isp thoda kam.

Source: UNIT 4 RP - L2.pdf | Page 16 se 18 (6 sources: 1-2 p16, 3 p17, 4-6 p18)

Pressurizing gas ke 6 sources

- **High-pressure inert gases** jo ambient temperature par stored hon. Sabse common. Typical gases: helium, nitrogen aur air.
- **Heated high-pressure inert gases** (usually 200 se 800 F ya 93 se 427 C). Isse required gas aur isliye pressurizing system ka inert mass kam hota hai.
- **Chemical reaction se bani gas** (liquid bipropellants ya monopropellant ya solid propellant se), aise mixture ratio par jisse 'warm gas' bane. Chemically generated warm gas wale tank pressurization systems usually heated inert gas systems se halke hote hain, khaas kar high total impulse applications me.
- **Cryogenic liquid propellant ka evaporation:** usually liquid hydrogen ya liquid oxygen ka chhota sa hissa evaporate karte hain, heat thrust chamber ke cooling jacket se ya turbine exhaust gases se (heat exchangers ke saath) le kar. Phir is evaporated flow ka kuch ya poora hissa tank pressurization ke liye use hota hai.
- **Direct injection:** chhoti hypergolic fuel stream ko main oxidizer tank me aur chhoti hypergolic oxidizer stream ko fuel tank me inject karna. Kai countries ne try kiya hai, par limited success mili.
- **Self-pressurization:** cryogenic propellants ko evaporation se khud pressurize karna. Feasible hai par control karna mushkil ho sakta hai.

Source: UNIT 4 RP - L2.pdf | Page 19 se 20

Pressurizing system design karne ke liye jaruri information

Koi bhi pressurizing system design ya analyze karne ke liye tank aur engine ki ye information chahiye hoti hai (inme se kai ki nominal, maximum aur minimum values chahiye):

- Propellant flow, thrust, duration, pulse width
- Propellant tank volume aur tank volume ka percent ullage (tank me khali jagah jo gas ke liye rehti hai)
- Storage temperature range
- Propellant aur pressurizing gas ki properties
- Propellant tank pressure aur gas tank pressure
- Unavailable residual propellant ki matra (jo tank me reh jaata hai aur use nahi ho paata)

Problems - Liquid Rocket

Dhyan do

Repo ke Unit 4 slides me koi numerical nahi tha. Ye 4 problems maine exam-type practice ke liye banaye hain. Formulas wahi hain jo slides me thrust, characteristic velocity aur specific impulse ke liye diye gaye hain (UNIT 3 Lecture 2 Page 7-15, Lecture 3 Page 18-21). Liquid engine me mixture ratio O/F ka concept bhi aata hai.

Formulas

Kya nikalna hai	Formula
Thrust	$F = \dot{m}\, V_e = C_F\, p_c\, A_t$
Specific impulse	$I_{sp} = F / \dot{m}$ (N.s/kg). Seconds me: I_{sp} / g_0, $g_0 = 9.81$ m/s^2
Exhaust velocity	$V_e = C_F\, C^* = I_{sp}$ (N.s/kg)
Characteristic velocity	$C^* = p_c\, A_t / \dot{m}$
Propellant mass	$m_p = \dot{m}\, t_b = I_t / I_{sp}$, $I_t = F\, t_b$
Fuel aur oxidizer flow	$\dot{m}_o = \dot{m}\,(O/F) / (1 + O/F)$, $\dot{m}_f = \dot{m} / (1 + O/F)$
Tank volume	$V = m / \rho$
Blow-down tank (isothermal)	$p_1\, V_1 = p_2\, V_2$

Question 1: Thrust, specific impulse aur propellant mass (fuel aur oxidizer alag alag)

Ek liquid rocket engine 50 kN thrust 100 s tak deta hai. Uska specific impulse 3000 N.s/kg hai aur oxidizer-to-fuel mixture ratio O/F = 2.2 hai. Oxidizer ki density 1140 kg/m^3 aur fuel ki density 810 kg/m^3 hai. Total mass flow rate, exhaust velocity, total propellant mass, oxidizer aur fuel ka alag mass aur tank volumes nikalo.

Given data

- F = 50 kN = 50000 N
- I_{sp} = 3000 N.s/kg
- t_b = 100 s
- O/F = 2.2
- ρ_o = 1140 kg/m^3, ρ_f = 810 kg/m^3

Formula:

$$\dot{m} = \frac{F}{I_{sp}} \qquad m_p = \dot{m}\, t_b \qquad \dot{m}_o = \frac{\dot{m}\,(O/F)}{1+(O/F)} \qquad \dot{m}_f = \frac{\dot{m}}{1+(O/F)}$$

Jahan,

$\dot{m}$	=	Total propellant mass flow rate, kg/s
I_{sp}	=	Specific impulse, N.s/kg
m_p	=	Total propellant mass, kg
$\dot{m}_o,\ \dot{m}_f$	=	Oxidizer aur fuel ki mass flow rate, kg/s
O/F	=	Oxidizer-to-fuel mixture ratio

$$\dot{m} = \frac{50000}{3000} = 16.667 \text{ kg/s} \qquad V_e = I_{sp} = 3000 \text{ m/s}$$

$$m_p = 16.667 \times 100 = 1666.7 \text{ kg}$$

$$\dot{m}_o = \frac{16.667 \times 2.2}{1 + 2.2} = 11.458 \text{ kg/s} \qquad \dot{m}_f = \frac{16.667}{1 + 2.2} = 5.208 \text{ kg/s}$$

$$m_o = 11.458 \times 100 = 1145.8 \text{ kg} \qquad m_f = 5.208 \times 100 = 520.8 \text{ kg}$$

$$V_o = \frac{1145.8}{1140} = 1.005 \text{ m}^3 \qquad V_f = \frac{520.8}{810} = 0.643 \text{ m}^3$$

Final answer

- Mass flow rate = 16.67 kg/s
- Exhaust velocity = 3000 m/s (I_{sp} = 3000/9.81 = 305.8 s)
- Total propellant = 1666.7 kg (oxidizer 1145.8 kg + fuel 520.8 kg)
- Tank volumes: oxidizer 1.005 m^3, fuel 0.643 m^3 (ullage alag se add hoga)

Question 2: Chamber pressure se thrust, exhaust velocity aur specific impulse
Ek liquid engine ka chamber pressure 6 MPa, throat area 0.01 m^2, characteristic velocity 1800 m/s aur thrust coefficient 1.6 hai. Thrust, mass flow rate, exhaust velocity aur specific impulse nikalo.

Given data

- p_c = 6 MPa = 6 x 10^6 Pa
- A_t = 0.01 m^2
- C* = 1800 m/s
- C_F = 1.6

Formula:

$$F = C_F p_c A_t \qquad \dot{m} = \frac{p_c A_t}{C^*} \qquad V_e = C_F C^* \qquad I_{sp} = \frac{F}{\dot{m}}$$

Jahan,

F	=	Thrust, N
p_c	=	Chamber pressure, Pa
A_t	=	Throat area, m^2
C^*	=	Characteristic velocity, m/s
C_F	=	Thrust coefficient

$$F = 1.6 \times 6 \times 10^6 \times 0.01 = 96000 \text{ N}$$

$$\dot{m} = \frac{6 \times 10^6 \times 0.01}{1800} = \frac{60000}{1800} = 33.33 \text{ kg/s}$$

$$V_e = 1.6 \times 1800 = 2880 \text{ m/s} \qquad I_{sp} = \frac{96000}{33.33} = 2880 \text{ N.s/kg}$$

Final answer

- Thrust = 96 kN
- Mass flow rate = 33.33 kg/s
- Exhaust velocity = 2880 m/s
- I_{sp} = 2880 N.s/kg = 293.6 s

Question 3: Required chamber pressure aur propellant mass

Ek liquid engine ko 80 kN thrust 60 s tak dena hai. C* = 1750 m/s, C_F = 1.55 aur throat diameter 120 mm hai. Mass flow rate, chamber pressure, specific impulse aur total propellant mass nikalo.

Given data

- F = 80 kN = 80000 N
- t_b = 60 s
- C* = 1750 m/s
- C_F = 1.55
- d_t = 120 mm = 0.12 m

Formula:

$$\dot{m} = \frac{F}{C_F C^*} \qquad A_t = \frac{\pi d_t^2}{4} \qquad p_c = \frac{\dot{m} C^*}{A_t} \qquad I_{sp} = C_F C^* \qquad m_p = \dot{m} t_b$$

Jahan,

$\dot{m}$ = Mass flow rate, kg/s

A_t = Throat area, m^2

d_t = Throat diameter, m

p_c = Chamber pressure, Pa

$$\dot{m} = \frac{80000}{1.55 \times 1750} = \frac{80000}{2712.5} = 29.49 \text{ kg/s}$$

$$A_t = \frac{\pi (0.12)^2}{4} = 0.011310 \text{ m}^2$$

$$p_c = \frac{29.49 \times 1750}{0.011310} = 4.563 \times 10^6 \text{ Pa}$$

$$I_{sp} = 1.55 \times 1750 = 2712.5 \text{ N.s/kg} \qquad m_p = 29.49 \times 60 = 1769 \text{ kg}$$

Final answer

- Mass flow rate = 29.49 kg/s
- Chamber pressure = 4.56 MPa
- I_{sp} = 2712.5 N.s/kg (276.5 s)
- Propellant mass = 1769 kg

Check: F = C_F p_c A_t = 1.55 x 4.563 x 10^6 x 0.01131 = 80,000 N.

Question 4: Blow-down feed system ka tank pressure

Ek blow-down feed system ka propellant tank total volume 0.5 m^3 hai. Shuru me tank me gas ka volume (ullage) 30% hai aur gas pressure 3.0 MPa hai. Saara propellant nikalne ke baad tank pressure kitna hoga? (Gas ka expansion isothermal maano.)

Given data

- Total tank volume = 0.5 m^3
- Initial gas volume, V_1 = 30% of 0.5 = 0.15 m^3
- Initial pressure, p_1 = 3.0 MPa
- Final gas volume, V_2 = 0.5 m^3 (saara propellant nikal gaya)

Formula: isothermal expansion

$$p_1 V_1 = p_2 V_2 \qquad \Rightarrow \qquad p_2 = \frac{p_1 V_1}{V_2}$$

Jahan,

$p_1,\ V_1$	=	Shuru ka gas pressure aur volume
$p_2,\ V_2$	=	End ka gas pressure aur volume

$$p_2 = \frac{3.0 \times 0.15}{0.5}$$

$$\boxed{p_2 = 0.9 \text{ MPa}}$$

Matlab tank pressure 3.0 MPa se girkar 0.9 MPa ho gaya (3.33 guna kam). Isi wajah se blow-down system me thrust burn ke saath girta hai, aur regulated system me nahi (UNIT 4 RP - L2.pdf, Page 13-15).